

Multiomics Study Suggests Alternate RNA Translation More Pervasive Than Thought

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NEW YORK – A team led by researchers at Northeastern University and the Parallel Squared Technology Institute (PTI) has identified thousands of proteins that resulted from alternative RNA translation, including a number of stable, relatively abundant ones that may be linked both to normal biology and disease.

Detailed in [a paper](#) published last month in *Nature*, the work potentially expands the human proteome as well as the role of alternate RNA translation, said Nikolai Slavov, senior author on the study and professor of bioengineering at Northeastern as well as director of the PTI.

Slavov and his team did not necessarily set out to identify new protein forms, he noted. Rather, the project began as an effort exploring whether cancer cells contain a higher proportion of non-degraded mistranslated proteins than healthy cells — a hypothesis that Slavov said stemmed from the observation that cancer is associated with increased levels of mistranslation and poorer control of protein fidelity.

Looking into this question, the researchers "discovered rather surprisingly that in fact, many proteins resulting from alternate RNA decoding are rather highly abundant, with many appearing as abundant or more abundant than the corresponding proteins predicted by the genetic code," Slavov said.

Additionally, he noted, a number of these alternate protein forms "are very stable, which is also different from what we had expected."

Initially, Slavov and his team believed these protein identifications were most likely mistakes, and further validation work established that a number of them were, in fact, misidentifications. However, he said, "a surprisingly large set of these proteins withstood all of the challenges we threw at them."

To explore the matter further, the researchers analyzed matched proteomic, transcriptomic, and genomic data from 1,094 human samples, including six tumor types and 26 kinds of healthy tissue. For each patient, they built a database of predicted protein sequences based on in silico translations of their transcriptome. They then searched their mass spec data against these predictions using the MaxQuant ModifiComb tool, which can identify whether a particular mass spectrum matches a modified form of a peptide found in the sample.

The group identified roughly 9 million mass spectra to apparently modified peptides, including 124,000 peptides with mass shifts consistent with amino acid substitutions (AAS), indicating that they were possibly the products of alternative translation.

The researchers then ran these peptides, which they termed substituted amino acid peptides (SAAPs), through a series of data filters and analyses to validate that the detected peptide sequences did in fact

show amino acid substitutions and that the observed mass shifts were not due to events like post-translational modifications or genetic mutations. This left them with 8,746 AASs across 1,767 proteins, including 1,955 substitutions that they were able to localize with high confidence.

The researchers then measured the relative abundance of these SAAPs compared to their corresponding canonical peptides, quantifying the ratio of the two forms. Their measurements found that for roughly 360 proteins, the alternate form was the most abundant.

Slavov suggested that given the observed abundance of these protein products and their stability, it is perhaps incorrect to think of them as translation errors.

"I think that many of those alternate decodings are probably part of normal physiological regulation, and that we need to think of the genetic code in a more expanded, high-dimensional way," he said.

To look at possible associations between SAAPs and disease, the researchers compared the relative abundance of alternate peptide forms in cancer and healthy tissue. While ratios were similar across cancerous and healthy tissues, certain alternate forms showed significantly higher abundance in certain cancer types. Gene ontology analysis found that abundant SAAPs were associated with functions like gene expression, cellular organization, and signaling.

Gloria Sheynkman, an associate professor at the University of Virginia School of Medicine whose research is focused on proteoforms and protein variation, said that while researchers have observed alternate translational events before, the Northeastern group's work suggests they may be much more common and biologically relevant than previously thought.

"The big thing I took away is that finally, someone has given us a quantitative estimate of the capacity of this recoding and shown that it is pervasive and potentially functional," she said.

Sheynkman noted that while establishing gold standard evidence for the presence of these alternate forms would require additional experiments, such as mass spec analyses adding synthetic peptides to the SAAPs, the Slavov lab's approach to validating their findings was "very rigorous."

"Even with really stringent thresholds, they still found plenty of [SAAPs]," she said. "If you are interested in the question of just how prevalent this is, I think it's hard, but I think [the researchers] are getting to kind of within an order of magnitude."

Sheynkman said that she would be interested to see associations between SAAPs and other conditions like neurodegenerative disease and aging. She added that the study could also push researchers interested in the biology of translation to look more closely at whether there are more general principles governing when and why such alternate translation happens.

Slavov said he and his colleagues are pursuing a number of efforts to further explore the presence and role of SAAPs. From a technical perspective, they are working to develop methods both to identify more SAAPs and to identify them with higher confidence. For instance, he said, they look to use alternative mass spec fragmentation methods like electron-based dissociation as well as alternative digestion enzymes to improve their peptide sequence coverage.

They also aim to investigate the functional implications of SAAPs, including potentially by using AI structural models to make and test predictions about the impact of amino acid changes at the level of the individual proteins, as well as the level of protein interactions and cellular processes.

Slavov also suggested that emerging single-molecule protein sequencing technologies from companies like Quantum-Si could help with targeted validation of SAAPs.

"I see those methods as being less powerful than mass spectrometry, but to the extent that they are different, having an orthogonal way to confirm an unexpected sequence is super valuable," he said. "I think as a second, third step of independent validation, those methods can be really helpful, and I'll be excited to use them."

Sheynkman, who [has used](#) Quantum-Si's Platinum protein sequencing platform to detect protein variants produced by genetic processes like alternative splicing, likewise said that single-molecule approaches could be helpful for validating SAAPs and other variants.

"The proteome is so complex that we have to use multiple approaches," she said. "When you're in the proteoform paradigm, it's not just about sampling peptides or epitopes. You have to get at those distinguishing features, which are just a small sliver, but a very important sliver, of the proteome."

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